Problem 1

Redbull's test drive

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1 Problem

During a test run, a Formula 1 team used a modified physics engine in their simulation. You are provided with telemetry data from a car accelerating along a straight track. Your task is to analyze the data and infer the underlying physical behavior.

The data file `telemetry.json` contains the following recorded quantities:

- t – time (s)
- v – velocity (m/s)
- a – acceleration (m/s^2)
- F_{engine} – engine force (N)
- F_{drag} – drag force (N)
- m – mass (kg)

1.1 Tasks

1. **Consistency check** Examine whether the data is consistent with known physical laws. Specifically, test the relationship $F_{\text{net}} = ma$ using the provided forces. Also investigate the typical drag model $F_{\text{drag}} \propto v^2$. Summarize your findings.

2. **Identification of deviations.** Determine whether any standard physical assumptions appear to be violated. For example, does the drag force follow the expected quadratic dependence on velocity? Does the engine force behave as a constant or a function of velocity? Justify your reasoning using the data (e.g., plots, residuals, or quantitative comparisons).

3. **Model inference.** If deviations from standard physics are present, propose modified forms of the relevant relationships. Estimate any parameters required to describe these modifications (e.g., a different exponent for drag, a velocity-dependent engine force). Provide numerical values with appropriate units.

4. **Interpretation.** Discuss how the identified changes would affect the performance of the car at high speeds. Compare the predicted top speed and acceleration behavior to what would be expected under normal physics.

2 Appendix: Theoretical Background

2.1 Newton's second law

For a car moving along a straight line,

$$F_{\text{net}} = F_{\text{engine}} - F_{\text{drag}} = ma.$$

2.2 Standard drag model

At typical speeds, aerodynamic drag is modeled as

$$F_{\text{drag}} = \frac{1}{2}\rho C_d A v^2,$$

which simplifies to $F_{\text{drag}} = kv^2$ with k constant. Any deviation from the quadratic dependence may indicate a modified fluid behavior or a different physical regime.

2.3 Engine force

In conventional simulations, the engine force may be constant, velocity-limited (power-limited), or follow a specific torque curve. The data should reveal its actual dependence.